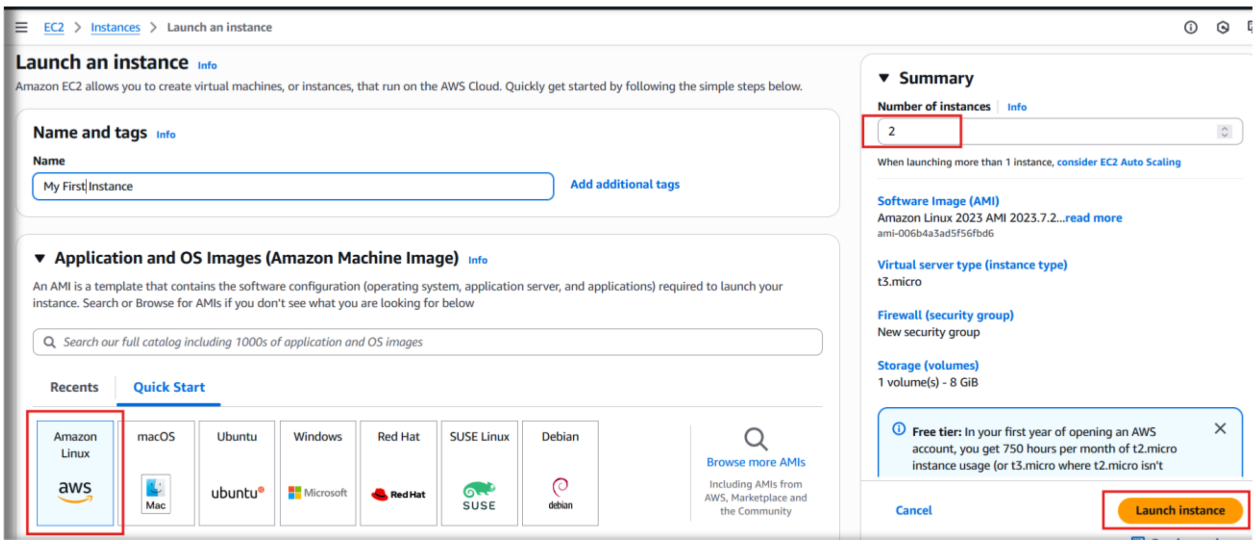
AWS Application Load Balancer (ALB)

🔰 What is a Load Balancer?

- An AWS Load Balancer automatically distributes incoming web traffic across multiple EC2 instances to ensure that no single server is overwhelmed.
- It improves the availability, scalability, and fault tolerance of your applications.
- Users access your website using the **DNS name** of the **Application Load Balancer (ALB)**.
- The **ALB receives the request** and checks which EC2 instance is **healthy and least loaded**.
- It then **routes the request** to that instance.

- If one EC2 instance fails, the **ALB automatically reroutes** traffic to healthy instances — ensuring **zero downtime** for users.

Step 1. Create 2 EC2 Instances



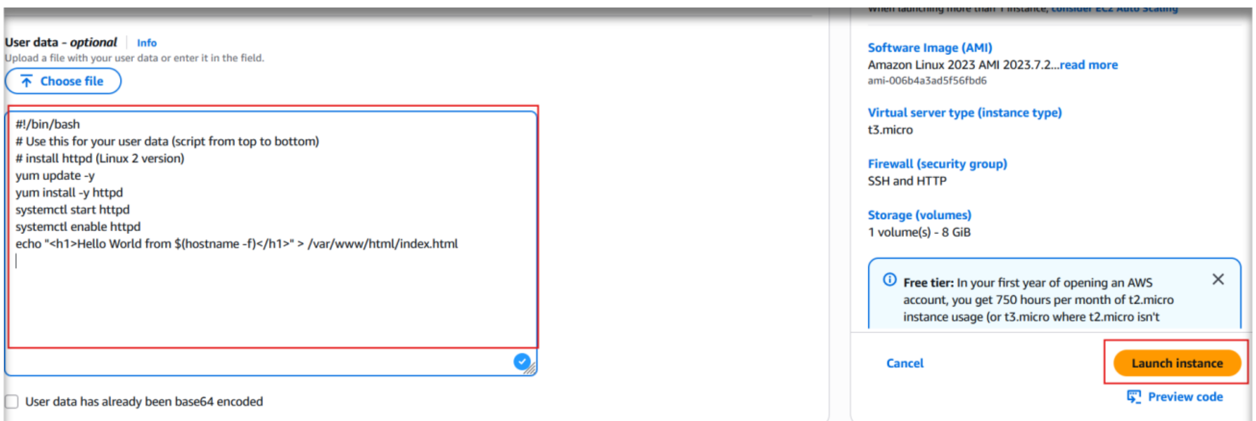
Note:

- **Add the port no 80 and 22 in security group** and
- Also fill **“User Data”** field which is used to provide a **script that will run**

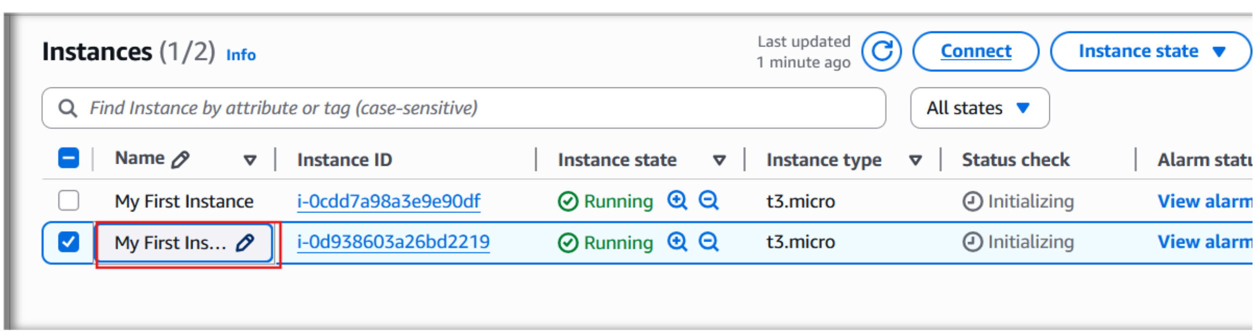
automatically when the instance is launched for the first time.

This process is called: **Bootstrapping**

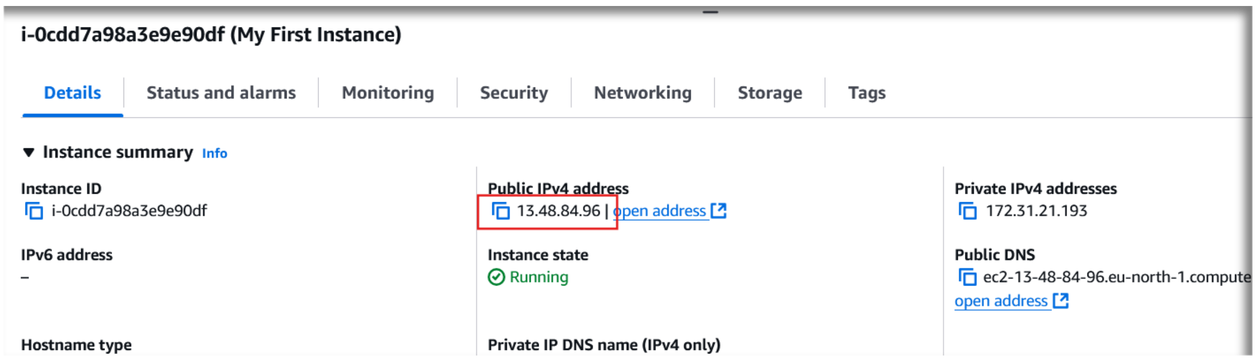
Lastly Launch Instance >

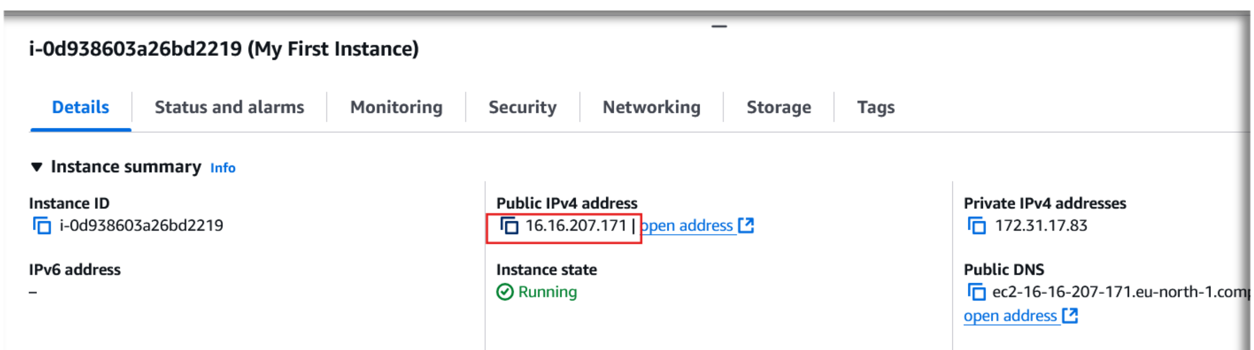


2 Instance got created, now rename second instance name **"My Second Instance".**

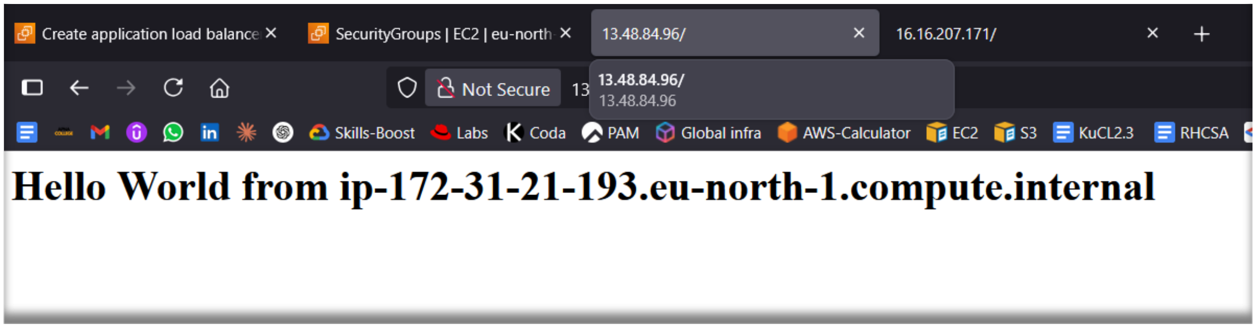


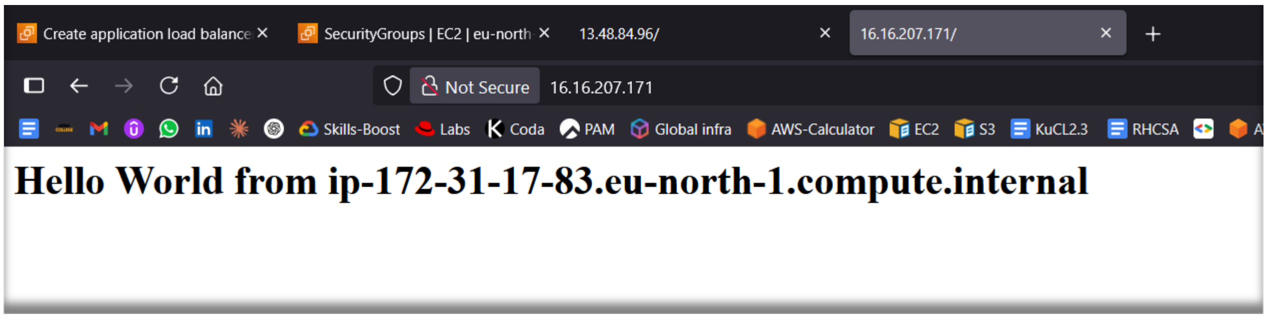
Copy the Ip addresses of both instances and paste in 2 different browser tabs.





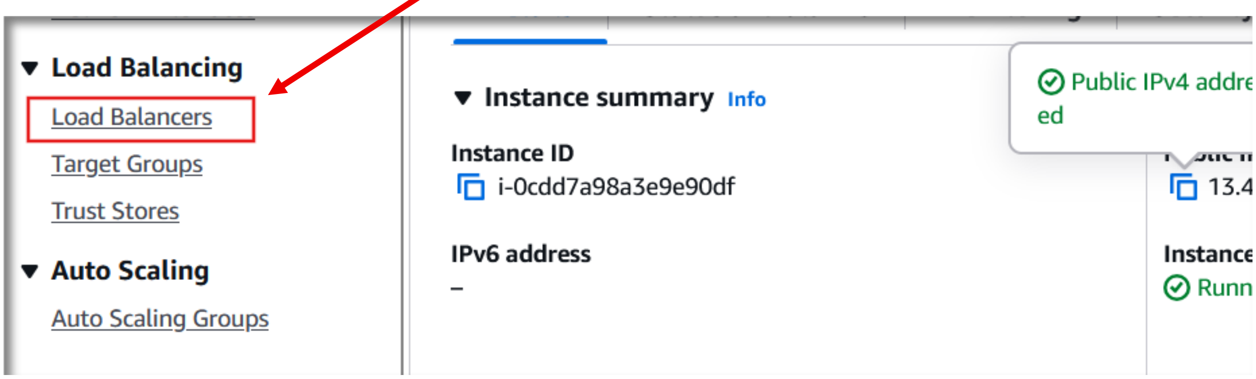
- You can see the instance is on 2 different browser tabs.
- It do not act like Load balancing on single tab.



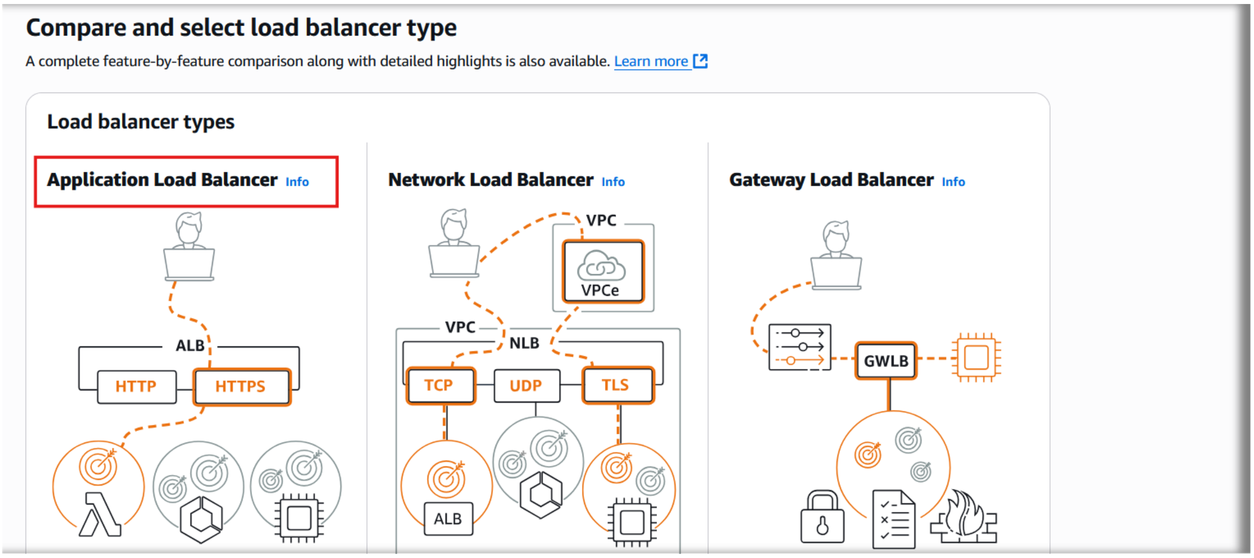


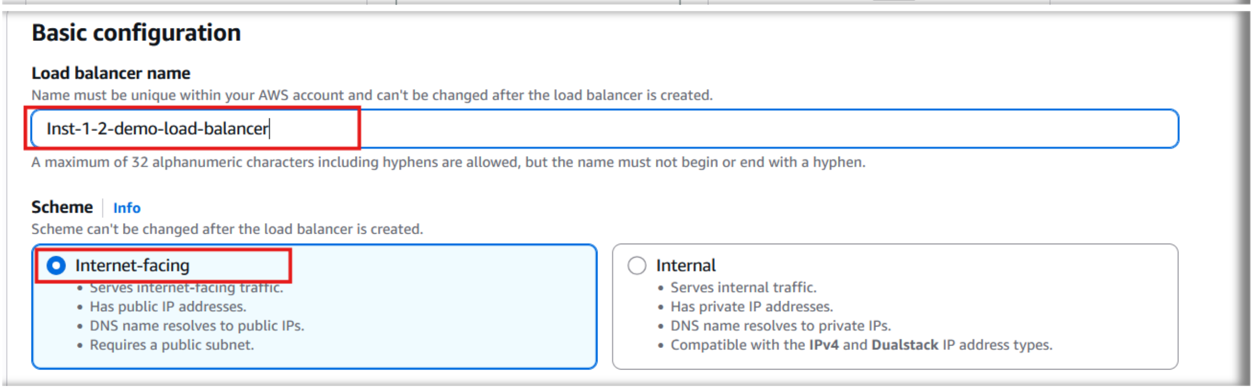
Here is the example of website hosting using AWS instance.

Step 2. Go to EC2 Load Balancing and create Load Balancer.

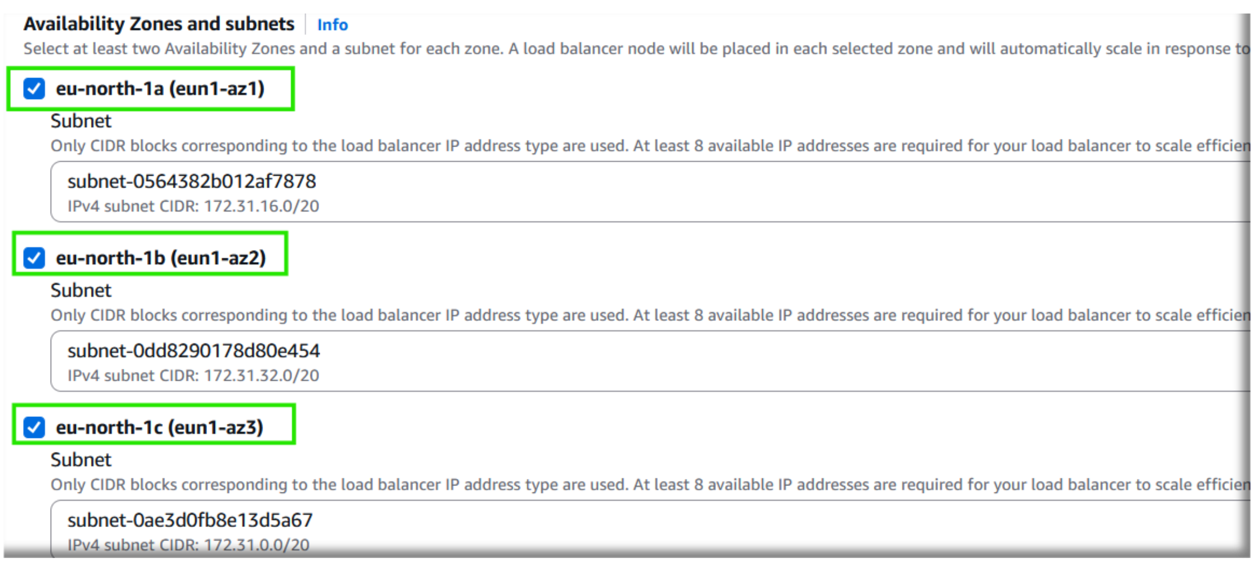


Create the Application Load Balancer

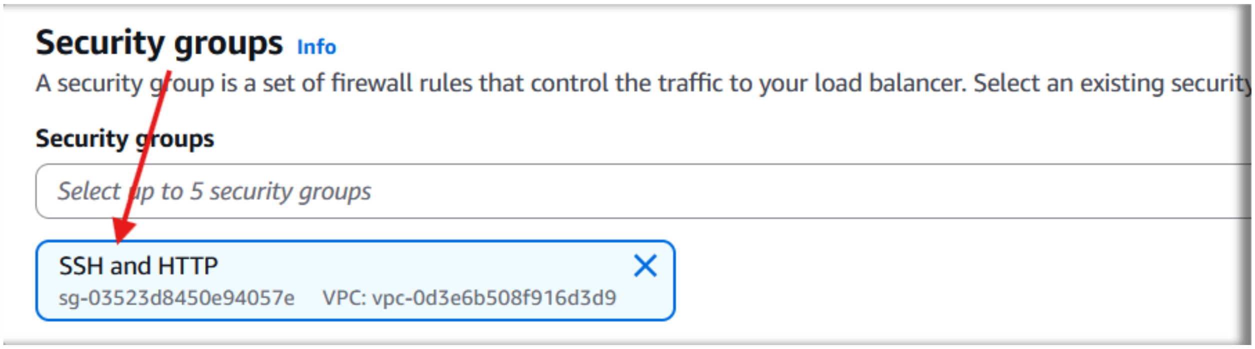




Select All the availability zones.

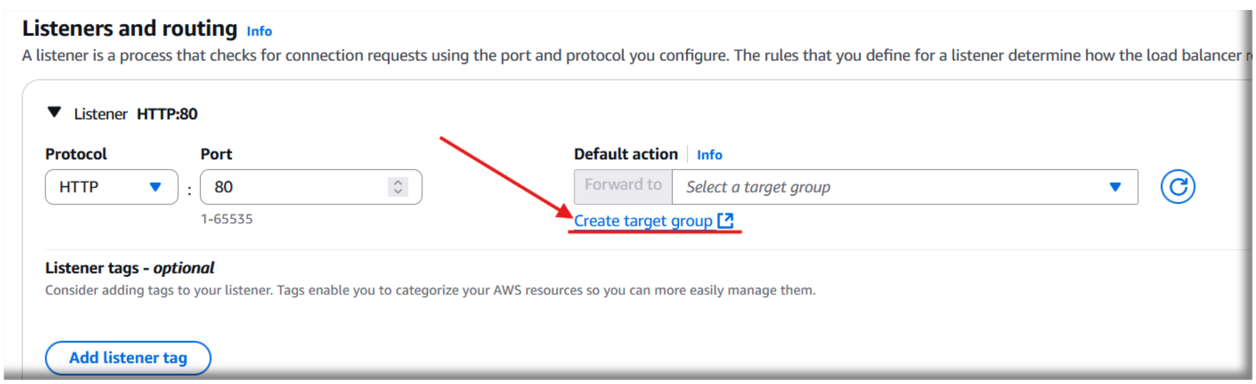


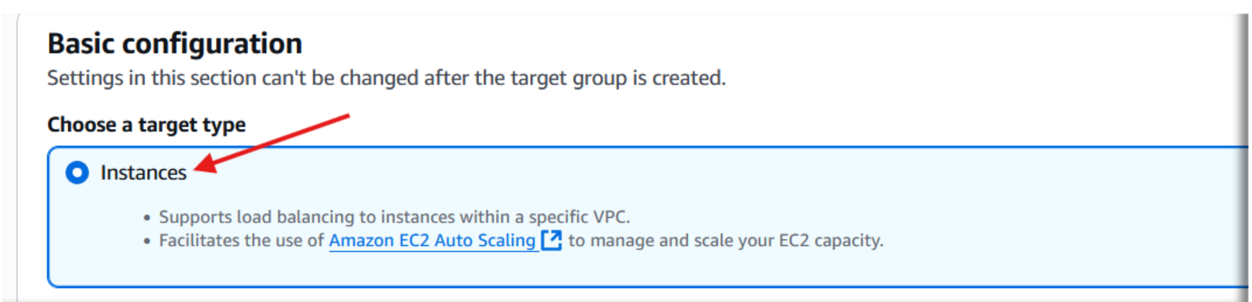
Enable the port number 80 and 22 in Security group.

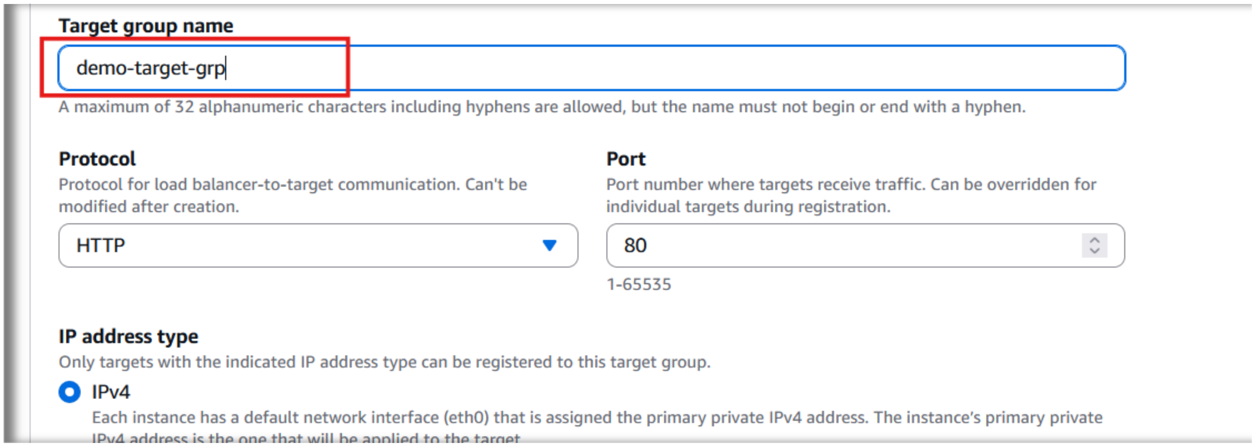


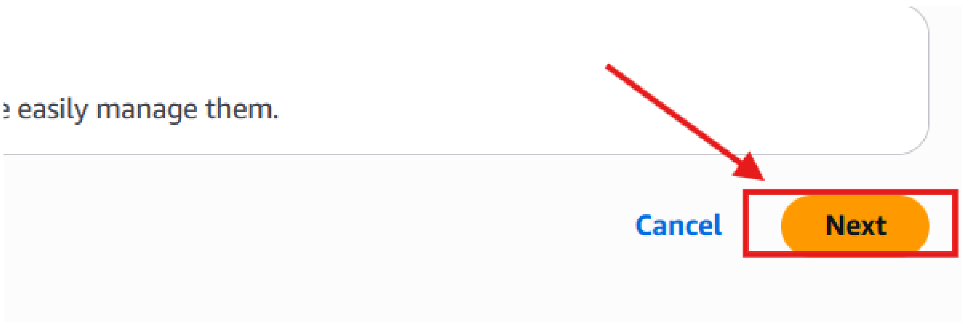
Now Create the Target Group.

Target Group is a logical grouping of backend resources (like EC2 instances) that a Load Balancer sends traffic to.

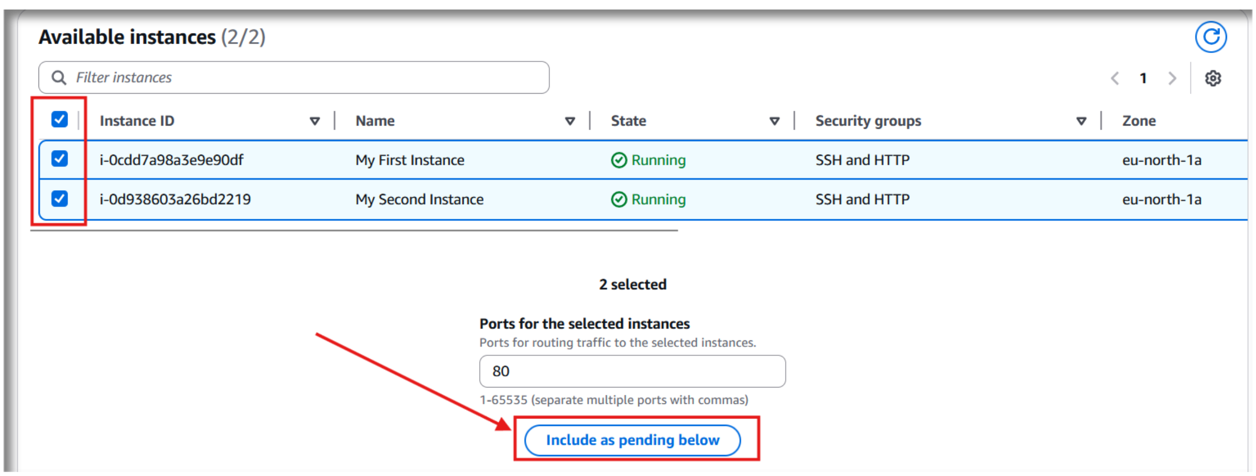


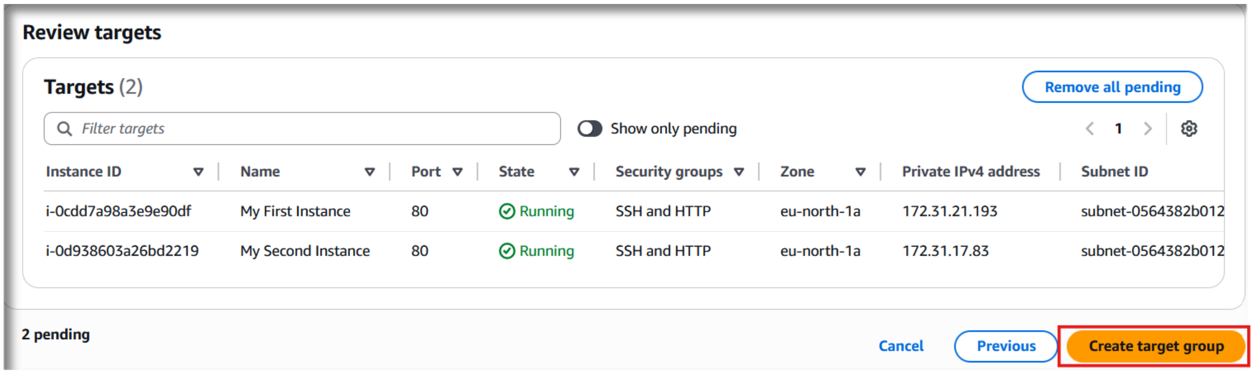


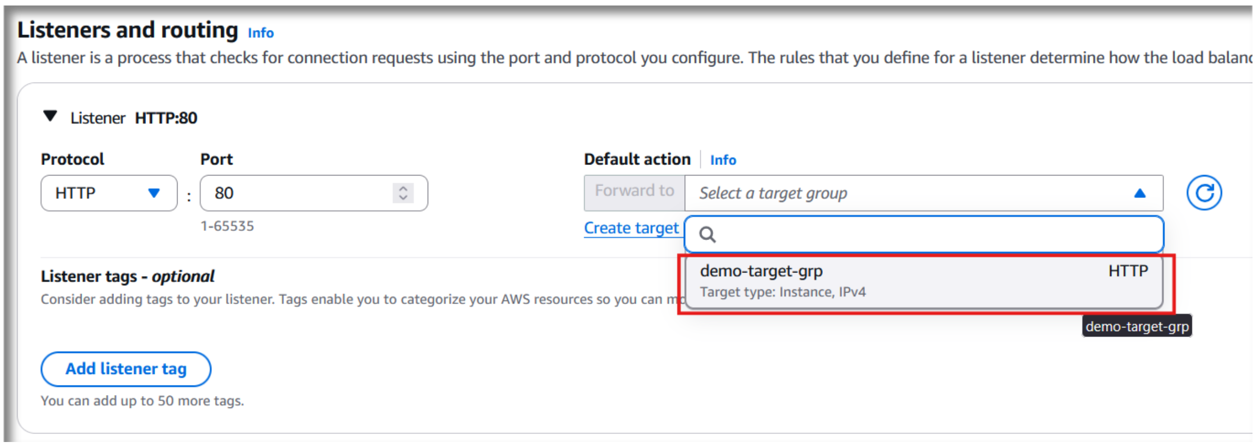




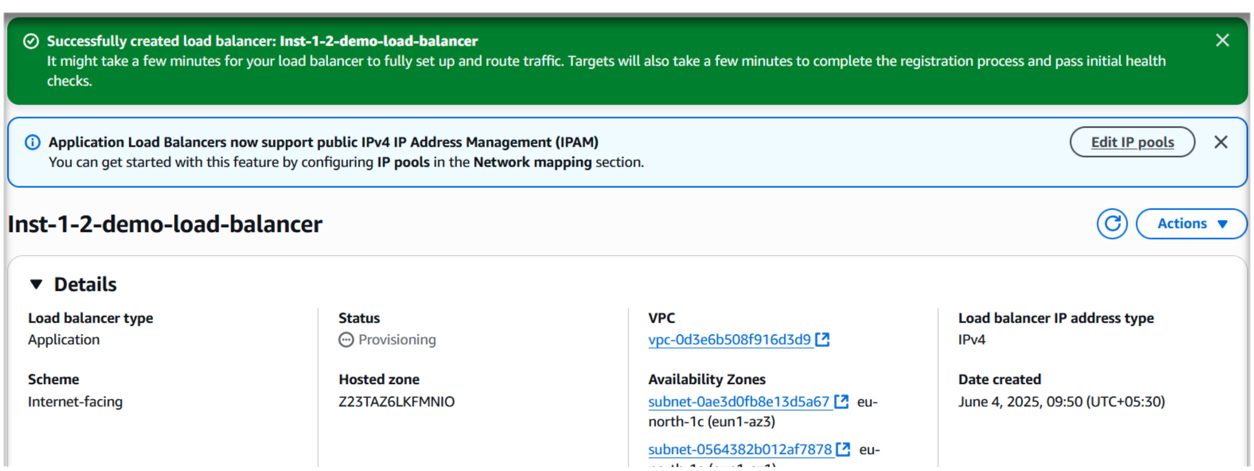
Select the instances to make target groups

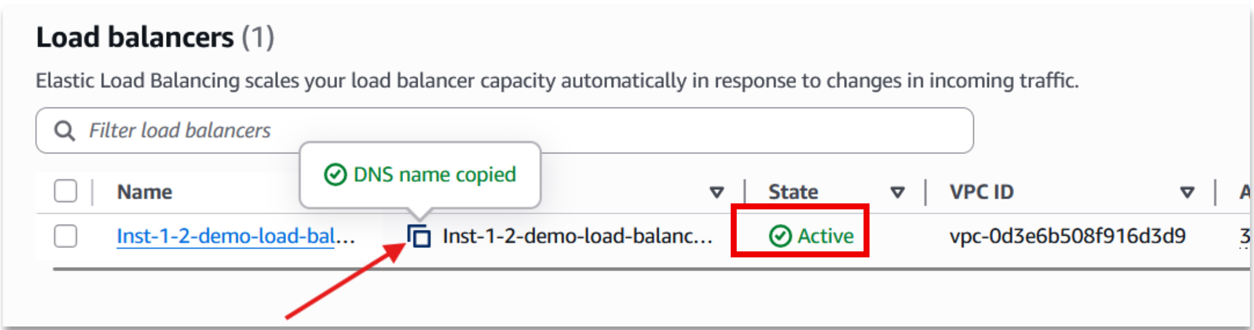






Target group is created, now click on create the Load Balancer.





✅ Conclusion:

Through this hands-on setup, I learned:

- 🔄 How traffic is intelligently distributed across multiple EC2 instances
- ▯ The importance of Target Groups and Health Checks for backend reliability
- ⚙️ Practical deployment flow involving EC2, Security Groups, ALB, and DNS
- 📶 Real-time performance benefits and automatic failover capability of ALB

Load Balancer is not just a tool—it's the backbone of reliability in large-scale web systems.